

Interpreting Accented Speech in Foundation Models via Sparse Autoencoders

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Abstract

This work aims to interpret how **accent information is encoded** in various speech representations. We propose to use **sparse autoencoders** to investigate whether specific dimensions within these representations capture the majority of accent-related information. This study not only provides insights into how **speech foundation models encode accent information** in their internal representations, but also facilitates **accent-related downstream applications**. For example, identifying salient dimensions may enable **dimension reduction**, leading to lower computational cost while improving model explainability.

Illustration

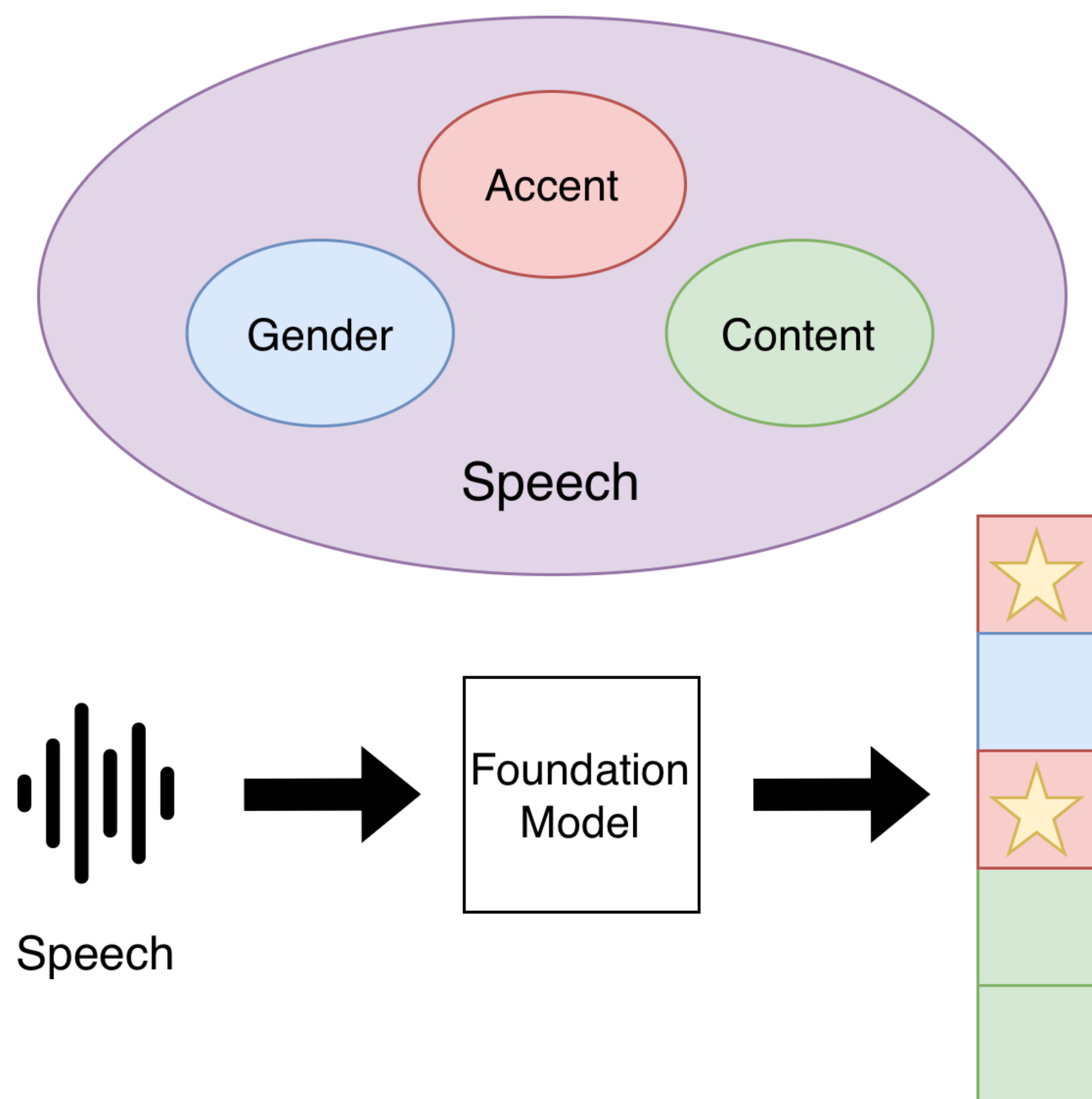
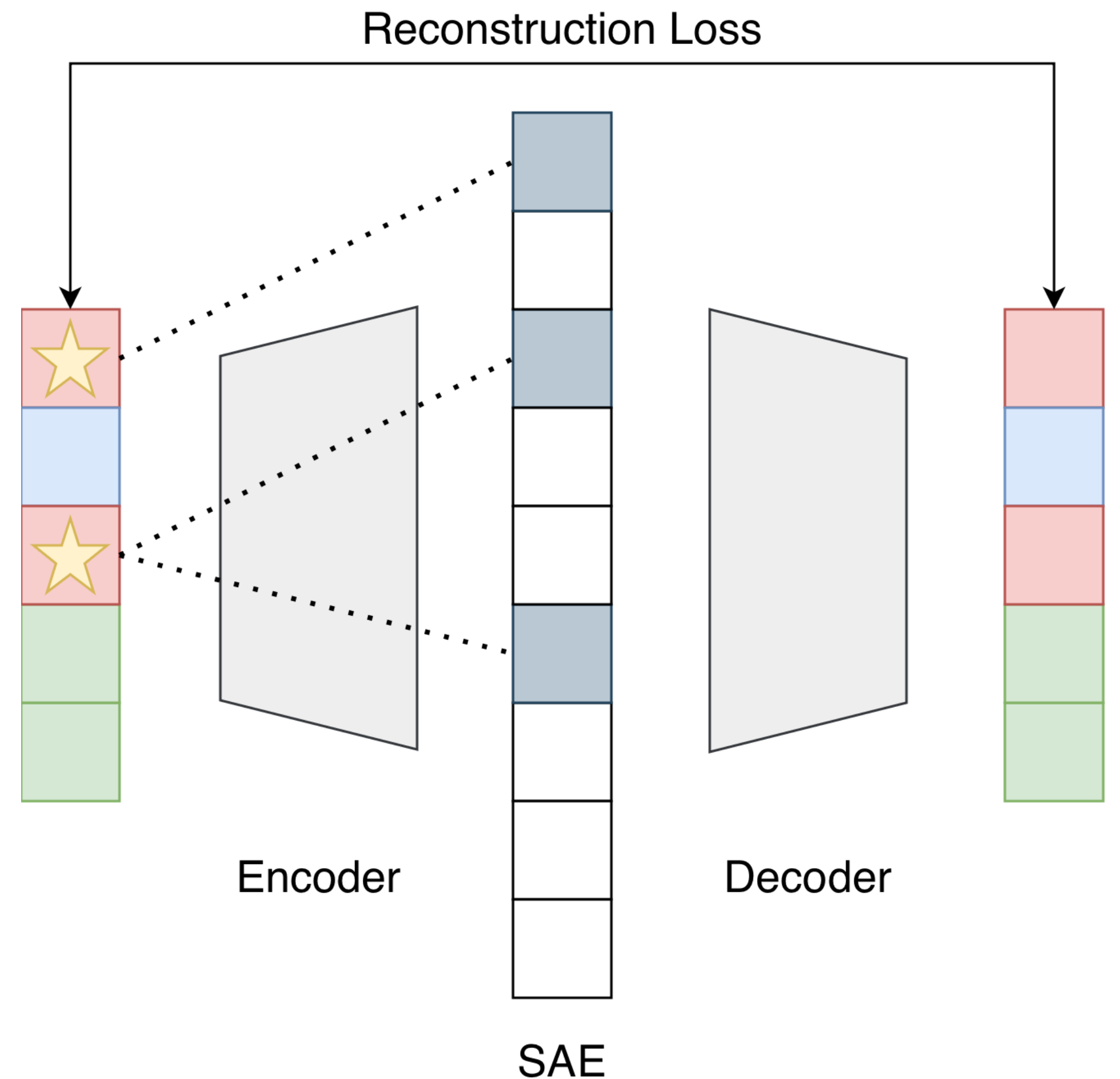


Fig. 1. We aim to identify the dimensions that contain the most accent-related information from latent representations.

Method

We train **sparse autoencoders (SAEs)** to identify the dimensions that contain the most accent-related information at each time frame. Specifically, we construct paired representations from data with **the same content and speaker but different accents**. Under this setting, the differences between the paired representations primarily capture accent-related information. By training SAEs on these paired data, we can localize the dimensions that account for most of this accent-induced variation.

Sparse Autoencoders (SAEs)



Data

To construct such paired data, we adopt the approach described in [1], which uses a TTS model to generate speech with identical linguistic content and the same speaker but different accents. To further amplify accent differences, we modify the speaker embedding to strengthen the accent, at the cost of introducing additional noise. This enables a two-level study with varying degrees of accent strength.

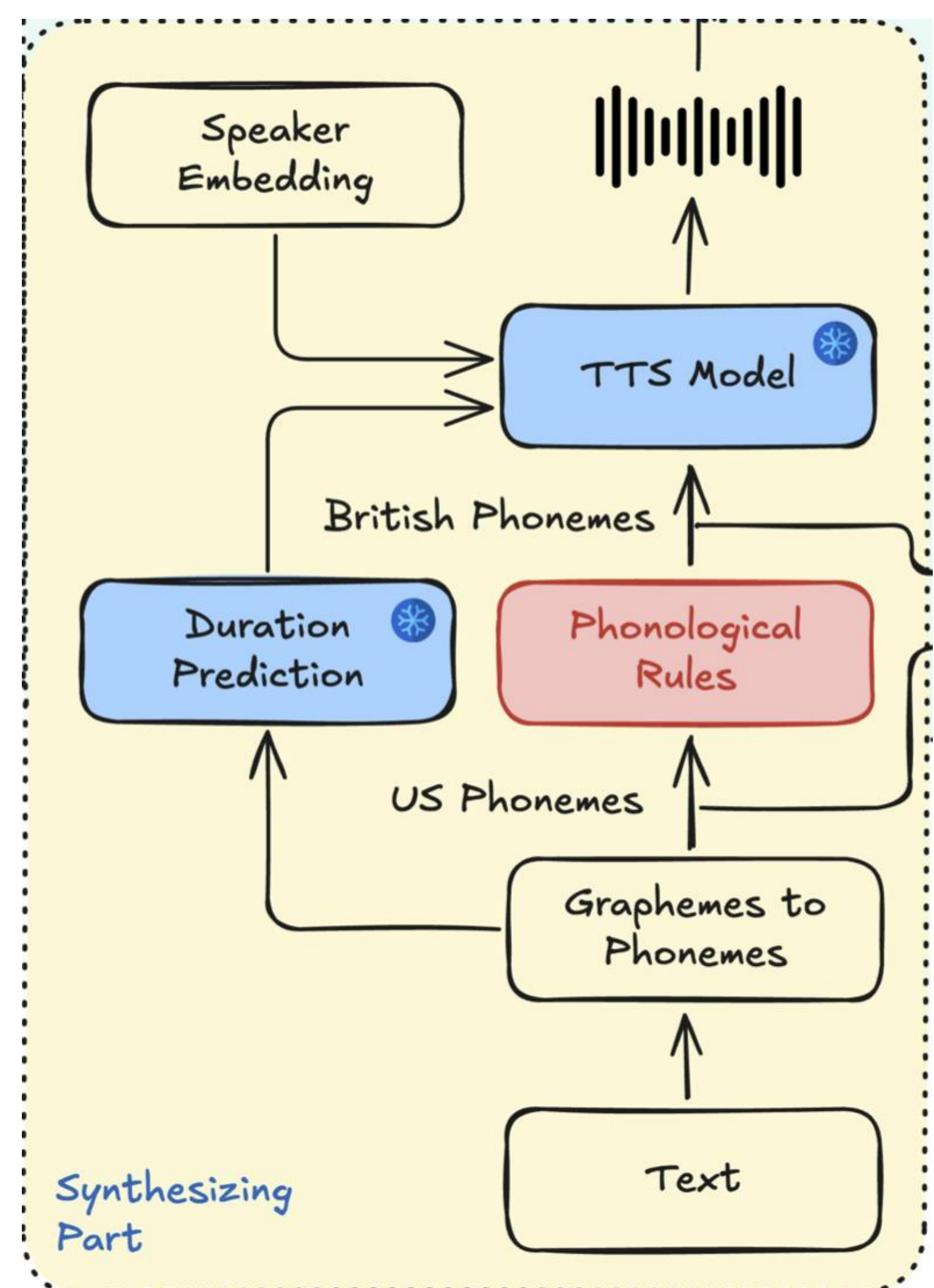


Fig. 2. Illustration of how to generate pair data (from [1]).